HW8

November 20, 2025

Exercise 1 Let $u \in \mathcal{C}^2([0,\infty) \times \mathbb{R})$ solve the wave equation in one dimension:

$$\begin{cases} u_{tt} - u_{xx} = 0, & (0,\infty) \times \mathbb{R}, \\ u = g,\ u_t = h, & \{t = 0\} \times \mathbb{R}. \end{cases}$$

Assume that g, h have compact support. Let

$$k(t) = \frac{1}{2}\int_{-\infty}^{\infty} |u_t(t,x)|^2\, dx, \quad p(t) = \frac{1}{2}\int_{-\infty}^{\infty} |u_x(t,x)|^2\, dx,$$

be the kinetic and potential energy.

1. Show that $k(t) + p(t)$ is constant in t (you do not have to use d'Alembert formula.)
2. Show that $p(t) = k(t)$ when t is large.

Exercise 2 Show that all *spherically symmetric* solutions $u(t,x) = u(t,|x|)$ of the 3d wave equation on the whole space

$$u_{tt}(t,x) = \Delta u(t,x), \quad t > 0,\ x \in \mathbb{R}^3,$$

can be written as

$$u(t,x) = \frac{F(|x| - t) + G(|x| + t)}{|x|}.$$

Hint: if $\varphi(x) = \varphi(|x|)$, *then* $\Delta\varphi(|x|) = \varphi''(r) + \frac{2}{r}\varphi'(r)$.

Exercise 3 Let $u \in \mathcal{C}^2(\mathbb{R}^4_+)$ solve

$$\begin{cases} u_{tt} - \Delta u = 0, & t > 0,\ x \in \mathbb{R}^3, \\ u(0,x) = \varphi(x), & x \in \mathbb{R}^3, \\ u_t(0,x) = \psi(x), & x \in \mathbb{R}^3, \end{cases}$$

with $\varphi, \psi \in \mathcal{C}_c^\infty(\mathbb{R}^3)$. Use the Kirchhoff's formula to show that there exists some constant $C > 0$ such that

$$|u(t,x)| \le \frac{C}{t}.$$

Exercise 4 Use separation of variables to solve

$$\begin{cases} u_{tt} - u_{xx} = 0, & x \in (0,\pi),\ t > 0, \\ u_x(t,0) = u_x(t,\pi) = 0, & t \ge 0, \\ u(0,x) = 0,\ u_t(0,x) = \cos x, & x \in [0,\pi]. \end{cases}$$

Exercise 5 Use separation of variables and Duhamel's principle to solve

$$\begin{cases} u_{tt} - u_{xx} = \sin x, & x \in (0, \pi),\ t > 0, \\ u(t, 0) = u(t, \pi) = 0, & t \geq 0, \\ u(0, x) = 0 = u_t(0, x) = 0, & x \in [0, \pi]. \end{cases}$$

Exercise 6 Use separation of variables to solve the following wave equation.

$$\begin{cases} u_{tt} - u_{xx} = 0, & x \in (0, \pi),\ t > 0, \\ u(0, x) = u_t(0, x) = 0, & x \in [0, \pi], \\ u_x(t, 0) = t,\ u_x(t, \pi) = 2t, & t \geq 0. \end{cases}$$

Hint: find a function $v(x)$ with $v'(0) = 1$, $v'(\pi) = 2$ and consider $u(t, x) - tv(x)$.

Ex 1. (1) $K'(t)+P'(t)=\int_{-\infty}^{\infty}(u_t u_{tt}+u_x u_{xt})\,dx$

$$=\int_{-\infty}^{\infty}(u_t u_{xx}+u_x u_{xt})\,dx$$

$$=(u_t u_x)\Big|_{-\infty}^{\infty}-\int_{-\infty}^{\infty}u_{tx}u_x\,dx+\int_{-\infty}^{\infty}u_x u_{xt}\,dx$$

注意到 g 和 h 的紧支性，故 $(u_t u_x)\Big|_{-\infty}^{\infty}=0$

从而 $K'(t)+P'(t)=0$，即得结论.

(2) Recall that 解 $u(t,x)=F(x+t)+G(x-t)$

其中 $F(x)=\frac{1}{2}g(x)+\frac{1}{2}\int_0^x h(y)\,dy$，$G(x)=\frac{1}{2}g(x)-\frac{1}{2}\int_0^x h(y)\,dy$

注意到，当 t 充分大时，左行波 F 与右行波 G 的支集不相交

此时对 $\forall x\in\mathbb{R}$，$F'(x+t)$ 和 $G'(x-t)$ 至少有一个为零.

故此时，$\int|u_t|^2dx=\int[G'(x-t)]^2dx+\int[F'(x+t)]^2dx-2\int F'(x+t)G'(x-t)\,dx$

$$=\int[G'(x-t)]^2dx+\int[F'(x+t)]^2dx+2\int F'(x+t)G'(x-t)\,dx=\int|u_x|^2dx$$

由此即得结论.

Ex 2. 由球对称得 $u_{tt}(t,r)=u_{rr}(t,r)+\frac{2}{r}u_r(t,r)$，其中 $r=|x|$

令 $v(t,r)=r\,u(t,r)$，代入得 $v_{tt}(t,r)=v_{rr}(t,r)$，$r>0$，即一维 wave equation

由 D'Alembert Formula 得解 $v(t,r)=F(t+r)+G(t-r)$

从而 $u(t,r)=\dfrac{v(t,r)}{r}=\dfrac{F(|x|+t)+G(|x|-t)}{|x|}$

Ex 3. 由 Kirchhoff 公式，$u(t,x)=\fint_{\partial B_t(x)}[t\psi(y)+\varphi(y)+\nabla\varphi(y)\cdot(y-x)]\,dS(y)$.

不失一般性，假设 $\mathrm{supp}(\varphi)$，$\mathrm{supp}(\psi)\subset D:=B_R(0)$

从而 $|u(t,x)|\le t\fint_{\partial B_t(x)}|\psi(y)|\,dS(y)+\fint_{\partial B_t(x)}|\varphi(y)|\,dS(y)+t\fint_{\partial B_t(x)}|\nabla\varphi(y)|\,dS(y)$

$$\le\frac{1}{4\pi t^2}\cdot|\partial B_t(x)\cap D|\cdot\left[t\cdot\|\psi\|_{L^\infty}+\|\varphi\|_{L^\infty}+t\cdot\|\nabla\varphi\|_{L^\infty}\right]$$

$$\le\frac{3R^2}{t}\cdot\max\{\|\psi\|_{L^\infty},\|\varphi\|_{L^\infty},\|\nabla\varphi\|_{L^\infty}\}$$

$$:=\frac{C}{t},\quad\forall t>0,\ x\in\mathbb{R}^3.$$

Ex 4. 由特征对得解应形如 $u(t,x)=\sum_{n=1}^{\infty}T_n(t)\cos(nx)$

代回方程得 $\begin{cases}T_n''+n^2T_n=0\\ T_n(0)=0,\ T_n'(0)=\begin{cases}1, & n=1\\ 0, & n\ge 2\end{cases}\end{cases}$

从而 $T_1(t)=\sin t$ 且 $T_n(t)=0$，$n\ge 2$

从而解为 $u(t,x)=\sin t\cos x$

Ex5. 由特征对得解应形如 $u(t,x)=\sum_{n=1}^{\infty} T_n(t)\sin(nx)$.

代回方程得 $\begin{cases} T_n''+n^2T_n=\begin{cases}1, & n=1\\ 0, & n\geq 2\end{cases} \\ T_n(0)=0,\ T_n'(0)=0 \end{cases}$

从而 $T_1(t)=\int_0^t \sin(t-s)\,ds = 1-\cos t$ 且 $T_n(t)=0,\ n\geq 2$

从而解为 $u(t,x)=(1-\cos t)\sin x$

Ex6. 令 $v(x)=x+\frac{x^2}{2\pi}$, $W(t,x)=u(t,x)-t\,v(x)$

从而 $\begin{cases} W_{tt}-W_{xx}=\frac{t}{\pi}, & x\in(0,\pi),\ t>0 \\ W_x(t,0)=0,\ W_x(t,\pi)=0 \\ W(0,x)=0,\ W_t(0,x)=-v(x)=-\left(x+\frac{x^2}{2\pi}\right) \end{cases}$

其解应形如 $W(t,x)=\sum_{n=0}^{\infty} T_n(t)\cos(nx)$

由 $\begin{cases} T_n''+n^2T_n=0 \\ T_n(0)=0,\ T_n'(0)=\frac{2}{\pi n^2}\left[1-2(-1)^n\right] \\ T_0(0)=0,\ T_0'(0)=-\frac{2}{3}\pi,\ T_0''(t)=\frac{t}{\pi} \end{cases}$ 得

$$u(t,x)=\frac{t^3}{6\pi}+t\left(x+\frac{x^2}{2\pi}-\frac{2\pi}{3}\right)+\frac{2}{\pi}\sum_{n=1}^{\infty}\frac{1-2(-1)^n}{n^3}\sin(nt)\cos(nx)$$